
hypotheses about model behavior. In contrast, Activation Oracle operates more like a black-box question-answering system: given activation vector(s), it produces natural language answers without exposing the underlying reasoning or mechanisms.

This trade-off has both advantages and disadvantages. The question-answer interface is simple, scalable, and potentially more accessible to practitioners who lack deep expertise in interpretability methods. However, it sacrifices the ability to drill down into mechanisms, verify hypotheses through targeted interventions, or build cumulative understanding of how models process information.

The model frequently makes incorrect guesses. An important limitation of Activation Oracles is reliability and lack of calibration. We observe the model making mistakes, and because it is not trained to express uncertainty, it will frequently produce an answer even when confidence is low.

Improving training data is a promising direction for future work. While we significantly expanded the diversity of training data compared to the original LatentQA, our datasets remain relatively small and limited in scope. We are particularly interested in self-supervised objectives that can scale by generating unlimited training data from pre-training corpora. Beyond self-supervised learning, we could also create better fine-tuning datasets with multi-turn interactions for user convenience, incorporate more diverse question types, collect “on-policy” generations for data from the original model, and potentially train via reinforcement learning. Combining scaled self-supervised objectives with these targeted improvements could address both the data scarcity and reliability issues.

Activation Oracles may not be the best approach for many practical problems. Our evaluation focuses primarily on auditing game benchmarks, where models are trained with specific hidden behaviors that cannot be easily detected through black-box prompting. While these benchmarks provide valuable controlled settings for comparing interpretability methods, they may not fully reflect many problems faced in real-world deployment. For instance, we found that Activation Oracle can identify hallucinations from single-token activations, but we also observed that simple prompting baselines achieve strong performance on hallucination detection in practice (see Appendix F for an anecdotal experiment). This suggests that for some common tasks, the additional complexity of activation-based interpretability may not provide sufficient advantage over simpler approaches.

That said, auditing benchmarks do capture scenarios that are particularly relevant for safety-critical applications, such as detecting intentionally concealed behaviors or models fine-tuned to deceive evaluation protocols. Understanding which applications truly benefit from methods like Activation Oracles remains an important direction for future research. This limitation is not unique to Activation Oracles, but rather reflects a common challenge facing many interpretability methods.

Our benchmarks are simplified and narrow. Our experiments use narrow fine-tunes that modify the base model along a small number of behavioral dimensions. Realistic post-training pipelines induce richer changes than the single-behavior fine-tunes we study. Our results therefore show that Activation Oracles can be effective in clean, controlled settings, but further work is needed to test them on more realistic post-trained systems. We view building more realistic models for model diffing and secret-keeping, and benchmarks that better capture the complexity of modern models, as important future work.

9 CONCLUSION

We demonstrate that training models to explain their own activation vectors with diverse tasks produces Activation Oracles that generalize to out-of-distribution settings and match or exceed state-of-the-art interpretability methods on downstream auditing benchmarks. By providing a simple natural-language interface to complex internal states, our approach offers a promising, accessible alternative to specialized interpretability techniques.

10 AUTHOR CONTRIBUTIONS

Adam Karvonen led the project, created the training infrastructure, implemented all experiments not otherwise mentioned, and wrote the initial draft of the paper with contributions from all authors. James Chua contributed to early exploratory experiments, trained the emergently misaligned models,

assisted with model diffing experiments, and revised the paper. Clément Dumas and Julian Minder implemented the emergent misalignment model diffing experiments. Kit Fraser-Taliente, Subhash Kantamneni, and Euan Ong implemented all experiments involving the Claude Haiku model. Arnab Sen Sharma and Daniel Wen created the classification datasets. Owain Evans and Samuel Marks jointly supervised the project and revised the paper. All authors contributed to discussions and provided feedback throughout the project.

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REFERENCES

- Valerio Basile, Johan Bos, Kilian Evang, and Noortje J. Venhuizen. Developing a large semantically annotated corpus. In *Proceedings of the Eighth International Conference on Language Resources and Evaluation (LREC'12)*, Istanbul, Turkey, 2012. European Language Resources Association (ELRA).
- Nora Belrose, Igor Ostrovsky, Lev McKinney, Zach Furman, Logan Smith, Danny Halawi, Stella Biderman, and Jacob Steinhardt. Eliciting latent predictions from transformers with the tuned lens, 2025. URL <https://arxiv.org/abs/2303.08112>.
- Jan Betley, Daniel Tan, Niels Warncke, Anna Szyber-Betley, Xuchan Bao, Martín Soto, Nathan Labenz, and Owain Evans. Emergent misalignment: Narrow finetuning can produce broadly misaligned llms, 2025. URL <https://arxiv.org/abs/2502.17424>.
- Samuel R. Bowman, Gabor Angeli, Christopher Potts, and Christopher D. Manning. A large annotated corpus for learning natural language inference. In Lluís Màrquez, Chris Callison-Burch, and Jian Su (eds.), *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*, pp. 632–642, Lisbon, Portugal, September 2015. Association for Computational Linguistics. doi: 10.18653/v1/D15-1075. URL <https://aclanthology.org/D15-1075/>.
- Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermy, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023. <https://transformer-circuits.pub/2023/monosemantic-features/index.html>.
- Bart Bussmann, Patrick Leask, and Neel Nanda. Batchtopk sparse autoencoders, 2024. URL <https://arxiv.org/abs/2412.06410>.
- Haozhe Chen, Carl Vondrick, and Chengzhi Mao. Selfie: Self-interpretation of large language model embeddings, 2024. URL <https://arxiv.org/abs/2403.10949>.
- Dami Choi, Vincent Huang, Sarah Schwettmann, and Jacob Steinhardt. Scalably extracting latent representations of users. <https://transluce.org/user-modeling>, 2025. Accessed: 2025-12-01.
- Anthony Costarelli, Mat Allen, and Severin Field. Meta-models: An architecture for decoding llm behaviors through interpreted embeddings and natural language, 2024. URL <https://arxiv.org/abs/2410.02472>.
- Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models, 2023. URL <https://arxiv.org/abs/2309.08600>.
- Bartosz Cywiński, Emil Ryd, Rowan Wang, Senthoran Rajamanoharan, Neel Nanda, Arthur Conmy, and Samuel Marks. Eliciting secret knowledge from language models, 2025. URL <https://arxiv.org/abs/2510.01070>.
- Emily Dinan, Angela Fan, Ledell Wu, Jason Weston, Douwe Kiela, and Adina Williams. Multi-dimensional gender bias classification. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 314–331, Online, November 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.emnlp-main.23. URL <https://www.aclweb.org/anthology/2020.emnlp-main.23>.
- Asma Ghandeharioun, Avi Caciularu, Adam Pearce, Lucas Dixon, and Mor Geva. Patchscopes: A unifying framework for inspecting hidden representations of language models, 2024. URL <https://arxiv.org/abs/2401.06102>.
- Avichal Goel, Yoon Kim, Nir Shavit, and Tony T. Wang. Learning to interpret weight differences in language models, 2025. URL <https://arxiv.org/abs/2510.05092>.

-
- Evan Hernandez, Arnab Sen Sharma, Tal Haklay, Kevin Meng, Martin Wattenberg, Jacob Andreas, Yonatan Belinkov, and David Bau. Linearity of relation decoding in transformer language models. In *Proceedings of the 2024 International Conference on Learning Representations*, 2024.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models, 2021. URL <https://arxiv.org/abs/2106.09685>.
- Subhash Kantamneni, Joshua Engels, Senthooan Rajamanoharan, Max Tegmark, and Neel Nanda. Are sparse autoencoders useful? a case study in sparse probing, 2025. URL <https://arxiv.org/abs/2502.16681>.
- Wan Hung Lee. English tense dataset. Kaggle, 2023. URL <https://www.kaggle.com/datasets/leewanhung/tense-dataset>.
- Belinda Z. Li, Zifan Carl Guo, Vincent Huang, Jacob Steinhardt, and Jacob Andreas. Training language models to explain their own computations, 2025a. URL <https://arxiv.org/abs/2511.08579>.
- Millicent Li, Alberto Mario Ceballos Arroyo, Giordano Rogers, Naomi Saphra, and Byron C. Wallace. Do natural language descriptions of model activations convey privileged information?, 2025b. URL <https://arxiv.org/abs/2509.13316>.
- Jack Lindsey. Emergent introspective awareness in large language models. *Transformer Circuits Thread*, 2025. URL <https://transformer-circuits.pub/2025/introspection/index.html>.
- Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets, 2024. URL <https://arxiv.org/abs/2310.06824>.
- Samuel Marks, Johannes Treutlein, Trenton Bricken, Jack Lindsey, Jonathan Marcus, Siddharth Mishra-Sharma, Daniel Ziegler, Emmanuel Ameisen, Joshua Batson, Tim Belonax, Samuel R. Bowman, Shan Carter, Brian Chen, Hoagy Cunningham, Carson Denison, Florian Dietz, Satvik Golechha, Akbir Khan, Jan Kirchner, Jan Leike, Austin Meek, Kei Nishimura-Gasparian, Euan Ong, Christopher Olah, Adam Pearce, Fabien Roger, Jeanne Salle, Andy Shih, Meg Tong, Drake Thomas, Kelley Rivoire, Adam Jermyn, Monte MacDiarmid, Tom Henighan, and Evan Hubinger. Auditing language models for hidden objectives, 2025. URL <https://arxiv.org/abs/2503.10965>.
- Julian Minder, Clément Dumas, Stewart Slocum, Helena Casademunt, Cameron Holmes, Robert West, and Neel Nanda. Narrow finetuning leaves clearly readable traces in activation differences, 2025. URL <https://arxiv.org/abs/2510.13900>.
- nostalgebraist. Interpreting gpt: The logit lens. <https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens>, 2020. LessWrong post.
- Alexander Pan, Lijie Chen, and Jacob Steinhardt. Latentqa: Teaching llms to decode activations into natural language, 2024. URL <https://arxiv.org/abs/2412.08686>.
- Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering llama 2 via contrastive activation addition, 2024. URL <https://arxiv.org/abs/2312.06681>.
- Gonalo Paulo, Alex Mallen, Caden Juang, and Nora Belrose. Automatically interpreting millions of features in large language models, 2025. URL <https://arxiv.org/abs/2410.13928>.
- Guilherme Penedo, Hynek Kydlíek, Loubna Ben allal, Anton Lozhkov, Margaret Mitchell, Colin Raffel, Leandro Von Werra, and Thomas Wolf. The fineweb datasets: Decanting the web for the finest text data at scale, 2024. URL <https://arxiv.org/abs/2406.17557>.
- Richard Socher, Alex Perelygin, Jean Wu, Jason Chuang, Christopher D. Manning, Andrew Ng, and Christopher Potts. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing*, pp. 1631–1642, Seattle, Washington, USA, October 2013. Association for Computational Linguistics. URL <https://www.aclweb.org/anthology/D13-1170>.